JIEMING ZHANG

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EDUCATION

University of Virginia **Charlottesville, VA**
B.S. Computer Engineering & B.S. Electrical Engineering, ***GPA: 3.9*** *Expected May 2029*
Relevant Coursework: Computer Systems Organization I, CPU Architecture, Digital Logic Design, Electric Circuits

TECHNICAL SKILLS

Languages: C++, Python, VHDL, Assembly, Java
Systems & Hardware: CPU Datapaths, Instruction Set Architecture, Digital Logic, Register Files, Stack Operations, Control Sequencing
Development Tools: Vim, Linux, Git, Quartus, ModelSim
Embedded Systems: ESP32, I2C, SPI, UART, CAN Bus, PCB Design

TECHNICAL PROJECTS

CPU Datapath **Spring 2026**
UVA ECE Coursework
- Built a 4-register × 8-bit VHDL toy CPU individually to course specification through component checkoffs, integrating ALU operations with immediate operands, load/store, and a conditional jump.
- Wrote testbenches and used ModelSim waveforms to find and fix opcode logic and testbench instruction errors; synthesized the CPU in Quartus.

Toy ISA Assembly Programming **Fall 2026**
Computer Systems Organization I | UVA
- Wrote Fibonacci, bitwise, loop, stack-operation, and arithmetic programs for an 8-bit ISA and translated hand-encoded hexadecimal programs into assembly using Vim on Linux.

IoT Golf Swing Analyzer & Visualizer Prototype **Summer 2026**
Embedded Hardware, Firmware & Python Analytics | Personal Project
- Built a first-prototype ESP32 PCB (~$20 components) with 3.7V LiPo-to-3.3V power management, BNO055 I2C, and MicroSD SPI interfaces.
- Implemented firmware that logs 100 Hz quaternion and accelerometer data to CSV and a Python pipeline that applies hand-tuned drift compensation, estimates clubhead velocity, and renders 3D swing trajectories.

ENGINEERING EXPERIENCE

Virginia Motorsports (Formula SAE) **Fall 2025 - Present**
Data Acquisition (DAQ) Subteam Member | University of Virginia
- Contributing to a redesign consolidating 5 PCBs into one centralized board handling 17 analog inputs across 4 sensor types.
- Designed schematics for the redesign and helped lay out the 2-layer spring 2026 prototype PCB in Altium and KiCad for sensor integration and data logging.
- Implemented an SD logging function that acquires a priority-based mutex and writes shared DAQ struct values to CSV on an SD card.

HACKATHON EXPERIENCE

VTHacks 14 **Sep 2026**
2-person team | Embedded Systems & Wireless Sensing
- Built ESP32 firmware that scanned 2.4 GHz Wi-Fi and Bluetooth traffic for packet counts and RSSI, storing an empty-room baseline in nonvolatile memory.
- Added a command to set a packet-count anchor and linearly rescale the density score from that anchor.